WASP-1b

R-1402

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-1_150107` · created 2026-08-01T04:15:59+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457029.7 +/- 0.006 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.117 +/- 0.03
Transit depth (Rp/Rs)^2	1.37 +/- 0.69 [%]
Orbital Inclination (inc)	89.9 +/- 1.43
Airmass coefficient 1 (a1)	1.008 +/- 0.016
Airmass coefficient 2 (a2)	-0.0032 +/- 0.0031
Scatter in the residuals of the lightcurve fit is	1.54 %
Best Comparison Star	#3 - [317, 214]
Optimal Aperture	3.55
Optimal Annulus	15.16
Transit Duration (day)	0.1551 +/- 0.0047

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.117 ± 0.03 vs 0.1036 (deviation 0.45σ)
Duration fit vs published	0.1551 d vs 0.1567 d (deviation 0.33σ)
Mid-time O–C	16.3 min
Depth SNR	3.9
Residual scatter	1.54 %

Light curve

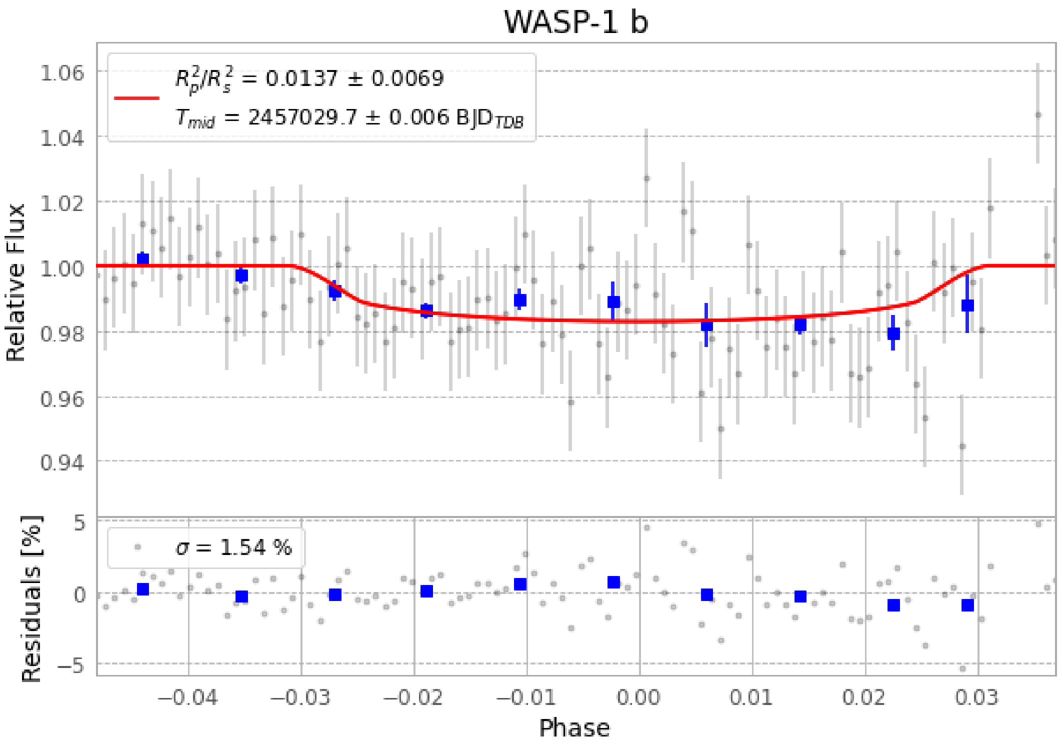

Figure 1 — FinalLightCurve_WASP-1 b_2015-01-06.png (SHA-256 9f167d10146ff3fa…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [310, 253], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 4653631fee237e58…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

108 FITS frames (71 MB), `WASP-1150107015112.FITS` → `WASP-1150107071212.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-1-150107.log (fef66bcc1a47…), FinalLightCurve_WASP-1 b_2015-01-06.png (9f167d10146f…), FinalLightCurve_WASP-1 b_2015-01-06.csv (641426136a41…)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-08-01 04:15Z.